

Workbreakdown Structure (WBS)

1. Project Initiation & Documentation

- 1.1 Define project objectives and goals
- 1.2 Identify key stakeholders and roles
- 1.3 Prepare initial project charter and scope outline
- 1.4 Documentation (maintaining all key project documents and updating them as needed throughout lifecycle)

2. Requirements Gathering & Analysis

- 2.1 Gather requirements from stakeholders
- 2.2 Identify functional requirements
- 2.3 Identify non-functional requirements (usability, availability, etc.)

3. Design

- 3.1 UI/UX Design (basic interface structure, initial layout decisions)
- 3.2 Database Design (ER diagram, schema notes)
- 3.3 Use Case Diagram (system interaction representation)

4. System Development

- 4.1 Student Interface Development
- 4.2 Login & Authentication
- 4.3 Admin Dashboard Development
- 4.4 Backend & Frontend Integration
- 4.5 WhatsApp Integration for canteen notifications

5. Testing & Quality Assurance

- 5.1 Unit Testing (testing each module individually)
- 5.2 Integration Testing (testing how modules work together — e.g., login + student interface + database)
- 5.3 System Testing (end-to-end testing of the entire application)
- 5.4 User Acceptance Testing (stakeholders test if the system meets needs)
- 5.5 Test additional features after implementation

6. Enhancements & Improvements

- 6.1 Implement additional features (if time/resources allow)
- 6.2 Improve UI based on testing and feedback
- 6.3 Re-test enhancements

7. Deployment & Launch

- 7.1 Prepare deployment environment
- 7.2 Deploy final application
- 7.3 Official project launch